

MH1200 Linear Algebra I.

Solutions to the problem list for Week #3.

Problem 1:

Solve the following linear system by *Gauss-Jordan elimination* and back-substitution.

$$\begin{aligned} 10y - 4z + w &= 1 \\ x + 4y - z + w &= 2 \\ 3x + 2y + z + 2w &= 5 \\ -2x - 8y + 2z - 2w &= -4 \\ x - 6y + 3z &= 1 \end{aligned}$$

Solution

Gauss-Jordan Elimination reduces the augmented matrix to reduced row-echelon form as follows:

$$\begin{aligned} &\left(\begin{array}{cccc|c} 0 & 10 & -4 & 1 & 1 \\ 1 & 4 & -1 & 1 & 2 \\ 3 & 2 & 1 & 2 & 5 \\ -2 & -8 & 2 & -2 & -4 \\ 1 & -6 & 3 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cccc|c} 1 & 4 & -1 & 1 & 2 \\ 0 & 10 & -4 & 1 & 1 \\ 3 & 2 & 1 & 2 & 5 \\ -2 & -8 & 2 & -2 & -4 \\ 1 & -6 & 3 & 0 & 1 \end{array}\right) \xrightarrow{\begin{array}{l} R_3 - 3R_1 \\ R_4 + 2R_1 \\ R_5 - R_1 \end{array}} \left(\begin{array}{cccc|c} 1 & 4 & -1 & 1 & 2 \\ 0 & 10 & -4 & 1 & 1 \\ 0 & -10 & 4 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -10 & 4 & -1 & -1 \end{array}\right) \\ &\xrightarrow{\begin{array}{l} R_3 + R_2 \\ R_5 + R_2 \end{array}} \left(\begin{array}{cccc|c} 1 & 4 & -1 & 1 & 2 \\ 0 & 10 & -4 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{\frac{1}{10}R_2} \left(\begin{array}{cccc|c} 1 & 4 & -1 & 1 & 2 \\ 0 & 1 & -2/5 & 1/10 & 1/10 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{R_1 - 4R_2} \left(\begin{array}{cccc|c} 1 & 0 & 3/5 & 3/5 & 8/5 \\ 0 & 1 & -2/5 & 1/10 & 1/10 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \end{aligned}$$

In the final reduced row-echelon matrix, the 3rd and 4th columns do not contain the leading entries of any row. Therefore we take the corresponding variables, z and w , as the free parameters, called s and t below. Then the general solution is

$$\begin{aligned} x &= \frac{8}{5} - \frac{3}{5}s - \frac{3}{5}t \\ y &= \frac{1}{10} + \frac{2}{5}s - \frac{1}{10}t \\ z &= s \\ w &= t \quad s, t \in \mathbb{R} \end{aligned}$$

□

Problem 2:

Solve the following linear system by *Gauss-Jordan elimination* and back-substitution.

$$\begin{aligned} 5x_1 - 2x_2 + 6x_3 &= 0 \\ -2x_1 + x_2 + 3x_3 &= 1 \end{aligned}$$

Solution

$$\left(\begin{array}{ccc|c} 5 & -2 & 6 & 0 \\ -2 & 1 & 3 & 1 \end{array} \right) \xrightarrow{R_2 + \frac{2}{5}R_1} \left(\begin{array}{ccc|c} 5 & -2 & 6 & 0 \\ 0 & 1/5 & 27/5 & 1 \end{array} \right) \xrightarrow{\frac{1}{5}R_1, 5R_2} \left(\begin{array}{ccc|c} 1 & -2/5 & 6/5 & 0 \\ 0 & 1 & 27 & 5 \end{array} \right) \xrightarrow{R_1 + \frac{2}{5}R_2} \left(\begin{array}{ccc|c} 1 & 0 & 12 & 2 \\ 0 & 1 & 27 & 5 \end{array} \right)$$

The 3rd column in the reduced row-echelon matrix does not contain a leading entry of any row, so we take the corresponding variable, x_3 , to be the free parameter, called t below. Then the general solution is

$$\begin{aligned} x_1 &= 2 - 12t, \\ x_2 &= 5 - 27t, \\ x_3 &= t \quad t \in \mathbb{R} \end{aligned}$$

□

Problem 3:

(From a 2020 Midterm.)

Consider the following augmented matrix.

$$\left[\begin{array}{cccc|c} 0 & 1 & -2 & 1 & 1 \\ 4 & 1 & -6 & 1 & 3 \\ 2 & -1 & 0 & -1 & 0 \end{array} \right].$$

Apply Gauss-Jordan elimination to determine a row equivalent matrix in reduced row echelon form. Show all the steps in your working.

Solutions

To do Gauss-Jordan we begin with Gaussian elimination.

The first step is to do a row swap so the top left corner isn't zero. We begin by swapping R_1 and R_3 . We get:

$$\left[\begin{array}{cccc|c} 2 & -1 & 0 & -1 & 0 \\ 4 & 1 & -6 & 1 & 3 \\ 0 & 1 & -2 & 1 & 1 \end{array} \right].$$

Next we want to make the 4 into a 0, so we: $R_2 \rightarrow R_2 - 2R_1$. We get:

$$\left[\begin{array}{cccc|c} 2 & -1 & 0 & -1 & 0 \\ 0 & 3 & -6 & 3 & 3 \\ 0 & 1 & -2 & 1 & 1 \end{array} \right].$$

Next we want to make the entry in position (3,2) into a 0 so we do the row operation: $R_3 \rightarrow R_3 - \frac{1}{3}R_2$. We get:

$$\left[\begin{array}{cccc|c} 2 & -1 & 0 & -1 & 0 \\ 0 & 3 & -6 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

This is in Row Echelon form. We are asked to use Gauss-Jordan elimination, so we need to go further until it is in Reduced Row Echelon form. The first thing to do is to make the leading terms into +1. To do this we multiply R_1 by $1/2$ and R_2 by $1/3$. We get:

$$\left[\begin{array}{cccc|c} 1 & -1/2 & 0 & -1/2 & 0 \\ 0 & 1 & -2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

The final step to obtain Reduced Row Echelon form is to do row operations to make the columns with leading terms all zero (except the leading term itself). There is only one operation we need to do. $R_1 \rightarrow R_1 + \frac{1}{2}R_2$. We get:

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 0 & 1/2 \\ 0 & 1 & -2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

This matrix is in Reduced Row Echelon Form.

□

Problem 4:

We consider the following linear system.

$$\begin{aligned} x + y + az &= 0 \\ x + ay + z &= 0 \\ ax + y + z &= 0 \end{aligned}$$

Here a is a constant. Find the values of a for which the system has

- (i) no solution
- (ii) exactly one solution
- (iii) infinitely many solutions.

Solution

First we do Gaussian elimination to put the matrix Row echelon form. Notice we never have to divide by zero doing this so we don't have to split into subcases.

$$\left(\begin{array}{ccc|c} 1 & 1 & a & 0 \\ 1 & a & 1 & 0 \\ a & 1 & 1 & 0 \end{array} \right) \xrightarrow{\substack{R_2 - R_1 \\ R_3 - aR_1}} \left(\begin{array}{ccc|c} 1 & 1 & a & 0 \\ 0 & a-1 & 1-a & 0 \\ 0 & 1-a & 1-a^2 & 0 \end{array} \right) \xrightarrow{R_3 + R_2} \left(\begin{array}{ccc|c} 1 & 1 & a & 0 \\ 0 & a-1 & 1-a & 0 \\ 0 & 0 & 2-a-a^2 & 0 \end{array} \right)$$

Now we study this matrix. The shape of the matrix will change depending on what values a takes. The values of a where some of the entries become zero are:

$$a - 1 = 0 \quad \text{or} \quad 2 - a - a^2 = (a - 1)(-a - 2) = 0.$$

Thus we have to treat the cases $a = 1$ and $a = -2$ separately.

(Case $a = 1$.) In this case the matrix becomes:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

The system is consistent and there are two columns not containing leading entries. These columns give free parameters. Thus in this case there are infinitely many solutions.

(Case $a = -2$.) In this case the matrix becomes:

$$\left(\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Again, the system is consistent and there is a column not containing the leading entry of a row, so there will be an associated free parameter. In this case there are infinitely many solutions.

(Case $a \neq 1, -2$.) In this case the matrix looks like:

$$\left(\begin{array}{ccc|c} 1 & 1 & a & 0 \\ 0 & \neq 0 & \neq 0 & 0 \\ 0 & 0 & \neq 0 & 0 \end{array} \right)$$

In this case every column has the leading entry of some row, so there will be exactly one solution.

In summary:

- (i) There are no cases where there are no solutions. (This is not surprising given the system is homogeneous, so will always have the trivial solution $x = y = z = 0$.)
- (ii) When $a \neq 1, -2$ there is exactly one solution.
- (iii) When $a = 1$ or -2 there are infinitely many solutions.

□

Problem 5:

We consider the following linear system.

$$\begin{aligned} ax + 9y + 2z &= 2 \\ x + ay + 4z &= 4 \\ ay + 2z &= 0 \end{aligned}$$

Here a is a constant. Find the values of a for which the system has

- (i) no solution,
- (ii) exactly one solution,
- (iii) infinitely many solutions.

We'll solve this in two different ways. The first will just be a direct approach which will require a number of sub-cases. The second will start with a clever column swap, which will simplify the calculations.

1st solution – Direct method.

Solution – 1st way: To answer these questions we need to put the augmented matrix of the system into row-echelon form.

$$\begin{aligned} \left(\begin{array}{ccc|c} a & 9 & 2 & 2 \\ 1 & a & 4 & 4 \\ 0 & a & 2 & 0 \end{array} \right) & \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|c} 1 & a & 4 & 4 \\ a & 9 & 2 & 2 \\ 0 & a & 2 & 0 \end{array} \right) \xrightarrow{R_2 - aR_1} \left(\begin{array}{ccc|c} 1 & a & 4 & 4 \\ 0 & 9 - a^2 & 2 - 4a & 2 - 4a \\ 0 & a & 2 & 0 \end{array} \right) \\ & \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & a & 4 & 4 \\ 0 & a & 2 & 0 \\ 0 & 9 - a^2 & 2 - 4a & 2 - 4a \end{array} \right) \end{aligned}$$

Case $a = 0$:

$$\left(\begin{array}{ccc|c} 1 & 0 & 4 & 4 \\ 0 & 0 & 2 & 0 \\ 0 & 9 & 2 & 2 \end{array} \right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & 0 & 4 & 4 \\ 0 & 9 & 2 & 2 \\ 0 & 0 & 2 & 0 \end{array} \right)$$

Every column except the last contains the leading entry of some row, so there is exactly one solution in this case.

Case $a \neq 0$:

$$\left(\begin{array}{ccc|c} 1 & a & 4 & 4 \\ 0 & a & 2 & 0 \\ 0 & 9 - a^2 & 2 - 4a & 2 - 4a \end{array} \right) \xrightarrow{R_3 - \frac{9-a^2}{a}R_2} \left(\begin{array}{ccc|c} 1 & a & 4 & 4 \\ 0 & a & 2 & 0 \\ 0 & 0 & -\frac{2(a^2-a+9)}{a} & 2 - 4a \end{array} \right)$$

Since the quadratic polynomial $a^2 - a + 9$ has no roots it is non-zero for all a . Therefore each column except the last contains the leading entry of some row in this case, so there is exactly one solution.

To summarize, combining the results of the two cases, we have found that the linear system has exactly one solution for all values of a .

2nd solution – Starting with a column swap.

The problem can be solved more easily by performing a column interchange at the beginning to move the ‘ a ’ entries in the 2nd column to the 3rd column. This corresponds to interchanging the x_2 and x_3 variables in the linear system, and this will of course not affect the number of solutions. This should be followed by row interchanges to move the ‘ a ’ in the first column down to the bottom row.

$$\begin{aligned} \left(\begin{array}{ccc|c} a & 9 & 2 & 2 \\ 1 & a & 4 & 4 \\ 0 & a & 2 & 0 \end{array} \right) & \xrightarrow{x_2 \leftrightarrow x_3} \left(\begin{array}{ccc|c} a & 2 & 9 & 2 \\ 1 & 4 & a & 4 \\ 0 & 2 & a & 0 \end{array} \right) & \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|c} 1 & 4 & a & 4 \\ a & 2 & 9 & 2 \\ 0 & 2 & a & 0 \end{array} \right) & \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & 4 & a & 4 \\ 0 & 2 & a & 0 \\ a & 2 & 9 & 2 \end{array} \right) \\ & \xrightarrow{R_3 - aR_1} \left(\begin{array}{ccc|c} 1 & 4 & a & 4 \\ 0 & 2 & a & 0 \\ 0 & 2 - 4a & 9 - a^2 & 2 - 4a \end{array} \right) & \xrightarrow{R_3 - (1 - 2a)R_2} \left(\begin{array}{ccc|c} 1 & 4 & a & 4 \\ 0 & 2 & a & 0 \\ 0 & 0 & 9 - a + a^2 & 2 - 4a \end{array} \right) \end{aligned}$$

As noted earlier, $a^2 - a + 9$ is non-zero for all a , so each column except the last contains the leading entry of some row in this case. We conclude that there is exactly one solution for all values of a . This is the same result as found using the first method, but note that we did not need to separate into different cases this time.

□

Problem 6:

Determine the number of solutions of the following linear system in the variables x , y and z , depending on the value of an unknown constant a .

$$\begin{aligned}x + 3y + (2a^4 - 32)z &= 37 + 2a \\4y &= 28 \\2x + 6y + (5a^4 - 80)z &= 76 + 5a\end{aligned}$$

Solution.

The augmented matrix corresponding to this system is:

$$\left[\begin{array}{ccc|c} 1 & 3 & 2a^4 - 32 & 37 + 2a \\ 0 & 4 & 0 & 28 \\ 2 & 6 & 5a^4 - 80 & 76 + 5a \end{array} \right]$$

To determine the number of solutions we just need to do Gaussian elimination to put the matrix into row echelon form. The first move is: $R_3 \rightarrow R_3 - 2R_1$. We get:

$$\left[\begin{array}{ccc|c} 1 & 3 & 2a^4 - 32 & 37 + 2a \\ 0 & 4 & 0 & 28 \\ 0 & 0 & a^4 - 16 & 2 + a \end{array} \right]$$

Note that:

- $a^4 - 16 = 0$ if $a = 2$ or $a = -2$ and is non-zero at all other a .
- $2 + a = 0$ if $a = -2$ and is non-zero at all other a .

We have three cases to analyse:

- $a = 2$. In this case the third row is of the form $000|(\neq 0)$, so the system is inconsistent in this case.
- $a = -2$. In this case the matrix becomes:

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 33 \\ 0 & 4 & 0 & 28 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

This matrix has a column without a leading entry so the system has a free parameter, so it has infinitely many solutions.

- $a \neq \pm 2$. In this case the matrix is of the form:

$$\left[\begin{array}{ccc|c} 1 & 3 & \neq 0 & 37 + 2a \\ 0 & 4 & 0 & 28 \\ 0 & 0 & \neq 0 & \neq 0 \end{array} \right].$$

This is a matrix in row echelon form such that every column (except the special right hand column) has a leading entry. Thus there are no free parameters and the system has a unique solution.

□

Problem 7:

Consider the system

$$\begin{aligned} ax + by &= f \\ cx + dy &= g \end{aligned}.$$

Assume that $a \neq 0$. Give a condition on a, b, c, d, f, g for this system to have a unique solution.

The augmented matrix associated to this system is:

$$\left[\begin{array}{cc|c} a & b & f \\ c & d & g \end{array} \right]$$

Because we know that a is non-zero we can do the following row operation $R_2 \rightarrow R_2 - \frac{c}{a}R_1$. After this operation the matrix becomes

$$\left[\begin{array}{cc|c} a & b & f \\ 0 & d - \frac{bc}{a} & g - \frac{cf}{a} \end{array} \right]$$

To have a unique solution we need every column to contain the leading entry of some row. Thus we need:

$$d - \frac{bc}{a} \neq 0.$$

Because $a \neq 0$ this is equivalent to the equation:

$$ad - bc \neq 0.$$

(This is not surprising. Later when we study determinants we will recognise that this is the determinant of the 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$. When we understand determinants we will see that there is a way to solve this problem using determinants.)

Problem 8:

Find an example of a linear system with 3 variables and 4 equations that has

- (i) no solution,
- (ii) exactly one solution,
- (iii) infinitely many solutions with *two arbitrary parameters*.

Solutions

- (i) A simple example is

$$0 = 1$$

$$0 = 0$$

$$0 = 0$$

$$0 = 0$$

where the ‘0’ on the left-hand sides means $0x + 0y + 0z$. Another example is

$$x + y + z = 1$$

$$x + y + z = 2$$

$$x = 3$$

$$x = -1$$

Here the first and second pairs of equations individually have no solutions.

- (ii)

$$x = 1$$

$$y = 1$$

$$z = 1$$

$$x + y + z = 3$$

- (iii) A simple example is

$$x = 0$$

$$0 = 0$$

$$0 = 0$$

$$0 = 0$$

The solution of this linear system is $x = 0$ with the other variables y and z being arbitrary. (Once again, the ‘0’ on the left-hand sides means $0x + 0y + 0z$.) Another example is

$$x + y + z = 1$$

$$2x + 2y + 2z = 2$$

$$3x + 3y + 3z = 3$$

$$4x + 4y + 4z = 4$$

Each equation has the same solution set as the first equation $x + y + z = 1$, whose general solution clearly has 2 free parameters.

□

Problem 9:

A certain physical quantity f depends on another quantity x by

$$f(x) = ax^4 + bx^3 + cx^2 + dx + e$$

where a, b, c, d, e are unknown constants. To determine a, b, c, d, e the following measurements have been taken.

x	0	1	2	3	4
$f(x)$	1	1	3	31	133

Find the values of a, b, c, d, e .

Solutions

Note that we can find the constant e immediately, since $1 = f(0) = e$. Using this, the remaining measurements give 4 equations for the unknowns a, b, c, d :

$$\begin{aligned} a + b + c + d &= 0 \\ 16a + 8b + 4c + 2d &= 2 \\ 81a + 27b + 9c + 3d &= 30 \\ 256a + 64b + 16c + 4d &= 132 \end{aligned}$$

The linear system can be solved by Gaussian Elimination (or Gauss-Jordan Elimination), but there are a few simplifications that can be done first in this case to make the computation easier: Divide the 2nd, 3rd, and 4th equations by 2, 3, 4 respectively, then write the system as

$$\begin{aligned} d + c + b + a &= 0 \\ d + 2c + 4b + 8a &= 1 \\ d + 3c + 9b + 27a &= 10 \\ d + 4c + 16b + 64a &= 33 \end{aligned}$$

Gaussian Elimination reduces the augmented matrix of this system to a row-echelon matrix as follows:

$$\begin{aligned} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 8 & 1 \\ 1 & 3 & 9 & 27 & 10 \\ 1 & 4 & 16 & 64 & 33 \end{array} \right) &\longrightarrow \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 7 & 1 \\ 0 & 2 & 8 & 26 & 10 \\ 0 & 3 & 15 & 63 & 33 \end{array} \right) &\longrightarrow \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 7 & 1 \\ 0 & 0 & 2 & 12 & 8 \\ 0 & 0 & 6 & 42 & 30 \end{array} \right) &\longrightarrow \\ \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 7 & 1 \\ 0 & 0 & 2 & 12 & 8 \\ 0 & 0 & 0 & 6 & 6 \end{array} \right) &\longrightarrow \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 7 & 1 \\ 0 & 0 & 1 & 6 & 4 \\ 0 & 0 & 0 & 1 & 1 \end{array} \right) \end{aligned}$$

Solving the linear system corresponding to the final augmented matrix, we find

$$\begin{aligned} a &= 1, \\ b &= 4 - 6a = 4 - 6 = -2, \\ c &= 1 - 3b - 7a = 1 - 3(-2) - 7 = 0, \\ d &= -c - b - a = -0 - (-2) - 1 = 1 \end{aligned}$$

Recalling that $e = 1$ we conclude that $f(x) = x^4 - 2x^3 + x + 1$.

□

Problem 10:

Solve the following system of nonlinear (!) equations.

$$\begin{aligned}x^2 - y^2 + 2z^2 &= 6 \\ 2x^2 + 2y^2 - 5z^2 &= 3 \\ 2x^2 + 5y^2 + z^2 &= 9\end{aligned}$$

Solution

We'll start by writing the equations in terms of new variables $X = x^2$, $Y = y^2$ and $Z = z^2$ which turns the non-linear system we started with into a linear system:

$$\begin{aligned}X - Y + 2Z &= 6 \\ 2X + 2Y - 5Z &= 3 \\ 2X + 5Y + Z &= 9\end{aligned}$$

We apply Gaussian Elimination to its augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 6 \\ 2 & 2 & -5 & 3 \\ 2 & 5 & 1 & 9 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 1 & -1 & 2 & 6 \\ 0 & 4 & -9 & -9 \\ 0 & 7 & -3 & -3 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 1 & -1 & 2 & 6 \\ 0 & 4 & -9 & -9 \\ 0 & 0 & 51/4 & 51/4 \end{array} \right)$$

The linear system corresponding to the row-echelon matrix (the final augmented matrix above) is

$$\begin{aligned}X - Y + 2Z &= 6 \\ 4Y - 9Z &= -9 \\ \frac{51}{4}Z &= \frac{51}{4}\end{aligned}$$

Its solution is easily found by back-substitution; the result is $X = 4$, $Y = 0$, $Z = 1$. In terms of the original variables this is $x^2 = 4$, $y^2 = 0$, $z^2 = 1$.

Finally we have to write down all solutions of this system. This is simple enough we can just guess, but we have to make sure we get all possibilities with the signs. We end up with 4 solutions:

$$\{(2, 0, 1), (-2, 0, 1), (2, 0, -1), (-2, 0, -1)\} \subset \mathbb{R}^3.$$

Problem 11:

(3 marks.)

Consider the following system consisting of an ordered list of three linear equations in five variables x_1, x_2, x_3, x_4 and x_5

$$\begin{aligned}x_3 + x_5 &= 1 \\x_2 + x_4 + x_5 &= 1 \\x_1 + x_2 + x_3 + x_4 + x_5 &= 1\end{aligned}$$

- (i) Write down the augmented matrix for this system.
- (ii) Do elementary row operations to transform this matrix into row echelon form.
- (iii) Do further elementary row operations to transform this matrix into reduced row echelon form.
- (iv) From that matrix write down a parametrization for the set of solutions of this system.
- (v) If we swap two columns in an augmented matrix, how are the corresponding sets of solutions related?
- (vi) If I multiply every entry in the i -th column of an augmented matrix by a non-zero constant c , assuming the column is to the left of the partition, how are the sets of solutions corresponding to the two matrices related? (No justification is necessary.)
- (vii) If we add an extra column of zeroes on the far left of an augmented matrix, how are the sets of solutions corresponding to the two matrices related?

Solution to (i).

The augmented matrix is:

$$\left[\begin{array}{ccccc|c} 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{array} \right].$$

Solution to (ii).

A single row swap $R_1 \leftrightarrow R_3$ will put the matrix into row echelon form.

$$\left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array} \right].$$

Solution to (iii).

To transform this matrix into reduced row echelon form, we need to do row operations so that in every column containing a leading entry, every other entry is zero. First we fix column 3 by performing $R_1 \rightarrow R_1 - R_3$. This gives:

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array} \right].$$

Then we fix column 2 by performing $R_1 \rightarrow R_1 - R_2$. This gives:

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array} \right].$$

The matrix now in reduced row echelon form.

Solution to (iv).

Columns 4 and 5 do not contain leading entries. We set the corresponding variables to free parameters:

$$x_4 = r, \quad x_5 = s.$$

Then we solve equation 3 for x_3 :

$$x_3 + x_5 = 1 \Rightarrow x_3 = 1 - s.$$

Then we solve equation 2 for x_2 :

$$x_2 + x_4 + x_5 = 1 \Rightarrow x_2 = 1 - r - s.$$

Finally we solve equation 1 for x_1 :

$$x_1 - x_5 = -1 \Rightarrow x_1 = s - 1.$$

So our parameterization is:

$$\begin{cases} x_1 = s - 1 \\ x_2 = 1 - r - s \\ x_3 = 1 - s \\ x_4 = r \\ x_5 = s \end{cases}, \quad r, s \in \mathbb{R}.$$

Solution to (v).

If we swap two columns, the resulting set of solutions S' is obtained from the original set of solutions S by swapping the two entries corresponding to the swapped columns in every element of S .

Solution to (vi).

You can obtain the new set of solutions from the old set of solutions by taking every element and dividing the i -th entry of the vector by the constant c .

Solution to (vii).

To get the new set of solutions from the old set of solutions, take every vector from the old set, and adjoin an arbitrary real number to the left of the vector. An example is: $(1, -1, 0, 1, 2)$ might become $(7, 1, -1, 0, 1, 2)$. Take all possible vectors you can build in this way from vectors of the old set.

□

More abstract or challenging problems:

Problem 12:

(from a Midterm exam 2020.)

Consider the following two vectors in $\mathbb{R}^3$ depending on parameters a and b

$$\vec{v} = (a, 1, 1), \vec{w} = (1, 1, b).$$

From these vectors define the following subset $P_{a,b}$ of $\mathbb{R}^3$ which depends on the parameters a and b :

$$P_{a,b} = \{r\vec{v} + s\vec{w}; r, s \in \mathbb{R}\} \subset \mathbb{R}^3.$$

- (i) Consider the special case $a = 1, b = 2$. Give 5 different examples of vectors lying in $P_{1,2}$.
- (ii) For what values of the parameters a and b is the vector $(3, 3, 3)$ an element of this set $P_{a,b}$?
- (iii) State a geometric interpretation of these sets $P_{a,b}$.

Solution to (i).

Part (i) was a warm-up. Basically a test of being able to understand set theory notation as in the first week.

A correct answer might be for example:

$$\begin{aligned} 0 * (1, 1, 1) + 0 * (1, 1, 2) &= (0, 0, 0) \\ 0 * (1, 1, 1) + 1 * (1, 1, 2) &= (1, 1, 2) \\ 1 * (1, 1, 1) + 0 * (1, 1, 2) &= (1, 1, 1) \\ 1 * (1, 1, 1) + 1 * (1, 1, 2) &= (2, 2, 3) \\ 2 * (1, 1, 1) + 1 * (1, 1, 2) &= (3, 3, 4) \end{aligned}$$

Solution to (ii).

The vector $(3, 3, 3)$ will lie in $P_{a,b}$ if there exist an r and an s such that

$$r(a, 1, 1) + s(1, 1, b) = (3, 3, 3).$$

This is equivalent to the equation:

$$(ar + s, r + s, r + bs) = (3, 3, 3).$$

This is equivalent to the following system of 3 linear equations in the variables r and s depending on parameters a and b :

$$\begin{aligned} ar + s &= 3 \\ r + s &= 3 \\ r + bs &= 3. \end{aligned}$$

We need to understand whether this system is consistent (in other words, whether it has at least one solution) depending on the values of the parameters a and b .

The augmented matrix for this system is:

$$\left[\begin{array}{cc|c} a & 1 & 3 \\ 1 & 1 & 3 \\ 1 & b & 3 \end{array} \right].$$

Let's start with a row swap $R_1 \leftrightarrow R_2$ to move the a to the second line so we don't immediately have to divide by it when we do Gaussian elimination.

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ a & 1 & 3 \\ 1 & b & 3 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - aR_1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1-a & 3(1-a) \\ 1 & b & 3 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1-a & 3(1-a) \\ 0 & b-1 & 0 \end{array} \right].$$

Now Gaussian elimination tells us to use the $(1-a)$ entry to make the $(b-1)$ entry zero. Thus we are led to split this analysis into two cases depending on whether that entry is zero or non-zero. In other words, whether $a = 1$ or $a \neq 1$.

Case $a = 1$.

In this case the augmented matrix becomes

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & b-1 & 0 \end{array} \right]$$

The row echelon form of this matrix shows it is always consistent, no matter what b is. (You can quickly check the cases $b = 1$ and $b \neq 1$).

Case $a \neq 1$.

In this case the matrix is of the form:

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & \neq 0 & \neq 0 \\ 0 & b-1 & 0 \end{array} \right].$$

We need to treat two subcases: $b = 1$ and $b \neq 1$.

Subcase $b = 1$.

In this case the matrix becomes

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & \neq 0 & \neq 0 \\ 0 & 0 & 0 \end{array} \right].$$

This system is consistent.

The remaining case is:

Subcase $b \neq 1$.

In this case the matrix becomes:

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & \neq 0 & \neq 0 \\ 0 & \neq 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & \neq 0 & \neq 0 \\ 0 & 0 & \neq 0 \end{array} \right].$$

This case is inconsistent.

We can summarise the results of this analysis as follows: In order for the vector $(3, 3, 3)$ to lie in the set $P_{a,b}$ it is necessary for at least one of a or b to be 1.

Solution to (iii).

In the case $a = b = 1$ the set is a **line** in the direction $(1, 1, 1)$. (Note that in this case $\vec{v} = \vec{w}$.)

In every other case when the two vectors are not parallel, the set is a **plane** containing the vectors $\vec{v}$ and $\vec{w}$.

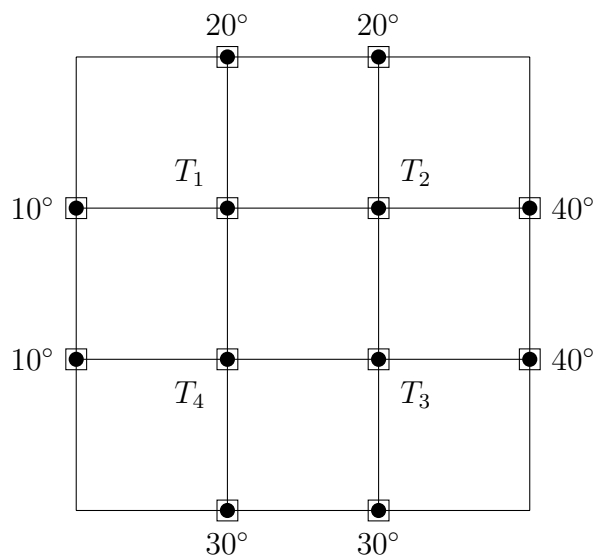
□

Comments from 2020 solutions.

The biggest problem with students responses here was imprecision. When a student writes: “ $a = 1$, $b = 1$ ”, do they mean they have to take those values simultaneously? Or is one condition enough? Also it wasn’t enough to guess the correct answer. To get full marks a student needed to give a completely clear analysis with considered all possible cases.

Problem 13:

[Determining temperatures] Imagine we have a metal plate and that we fix the temperature at the boundaries of this plate (see picture) and let the plate come to a thermal equilibrium. The temperature of a point in the interior of the plate can be approximated as the average of the temperature of the 4 neighboring points: above, below, left, and right. Set up and solve the linear equations giving the temperatures T_1, T_2, T_3, T_4 in the interior of the plate.



Solutions

This problem gives one application of solving systems of linear equations. Let's set up the system.

$$4T_1 - T_2 - T_4 = 30$$

$$4T_2 - T_1 - T_3 = 60$$

$$4T_3 - T_2 - T_4 = 70$$

$$4T_4 - T_1 - T_3 = 40$$

Putting it in augmented matrix form and doing Gaussian elimination gives

$$\begin{aligned} & \left[\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ -1 & 4 & -1 & 0 & 60 \\ 0 & -1 & 4 & -1 & 70 \\ -1 & 0 & -1 & 4 & 40 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & \frac{15}{4} & -1 & -\frac{1}{4} & \frac{135}{2} \\ 0 & -1 & 4 & -1 & 70 \\ 0 & -\frac{1}{4} & -1 & \frac{15}{4} & \frac{95}{2} \end{array} \right] \\ & \rightarrow \left[\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & \frac{15}{4} & -1 & -\frac{1}{4} & \frac{135}{2} \\ 0 & 0 & \frac{56}{15} & -\frac{16}{15} & 88 \\ 0 & 0 & -\frac{16}{15} & \frac{56}{15} & 52 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & \frac{15}{4} & -1 & -\frac{1}{4} & \frac{135}{2} \\ 0 & 0 & \frac{56}{15} & -\frac{16}{15} & 88 \\ 0 & 0 & 0 & \frac{24}{7} & \frac{540}{7} \end{array} \right] \end{aligned}$$

Now we solve by back substitution to find

$$T_4 = \frac{45}{2}, T_3 = 30, T_2 = \frac{55}{2}, T_1 = 20 .$$

□